AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 34 · TEACHER GUIDE

Test Diverse Users

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach coverage and its limits with complete records.

Keep datasets separate, supply four request cards and assess the untouched eight-case suite.

Keep one habit: Ask whether a failure is contractual, an unmet need or an untested condition before choosing the next check.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Evaluate across varied cases

Teach Week 34 as a two-session investigation using fictional data. A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall **pass rate** can hide that concentrated weakness.

Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can evaluate varied case types and prioritise an important failure slice.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of least privilege and tool scopes

Reference example, not the only acceptable wording: Teacher explanation: Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

No; they have become part of development **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Build the Recommender

Permissions and Human Review

Prompt Experiments

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

Concept to explain

Evaluation must include ordinary, edge, and **underserved** cases.

Term	Meaning
test profile	A description of the kinds of cases and conditions included in an evaluation.
edge case	An unusual, boundary or difficult input that can reveal a failure hidden by routine examples.
underserved	Receiving less adequate support or performance than the intended service requires.
fairness concern	A reason to examine whether a system creates unjustified differences in treatment or outcomes.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model category-level evidence before broad claims

> Keep six-case, twelve-case and user-condition traces separate.

State the six-case outcome contract

A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall **pass rate** can hide that concentrated weakness.

Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

Run the recorded-flag counter

```
checks = [{"category": "ordinary", "passed": True},
          {"category": "ordinary", "passed": True},
          {"category": "boundary", "passed": True},
          {"category": "invalid", "passed": True},
          {"category": "empty", "passed": True},
          {"category": "tie", "passed": False}]
passed = 0
for check in checks:
    if check["passed"]:
        passed = passed + 1
print(passed, len(checks))
```

The script reports 5 and 6. About 83.3 percent of these checks pass, but that is not an accuracy estimate for all future use.

The cases were deliberately chosen, their frequencies are artificial, and each category has few cases.

> **If ten copies of one passing ordinary case are added, what becomes the pass fraction? Does coverage improve?**

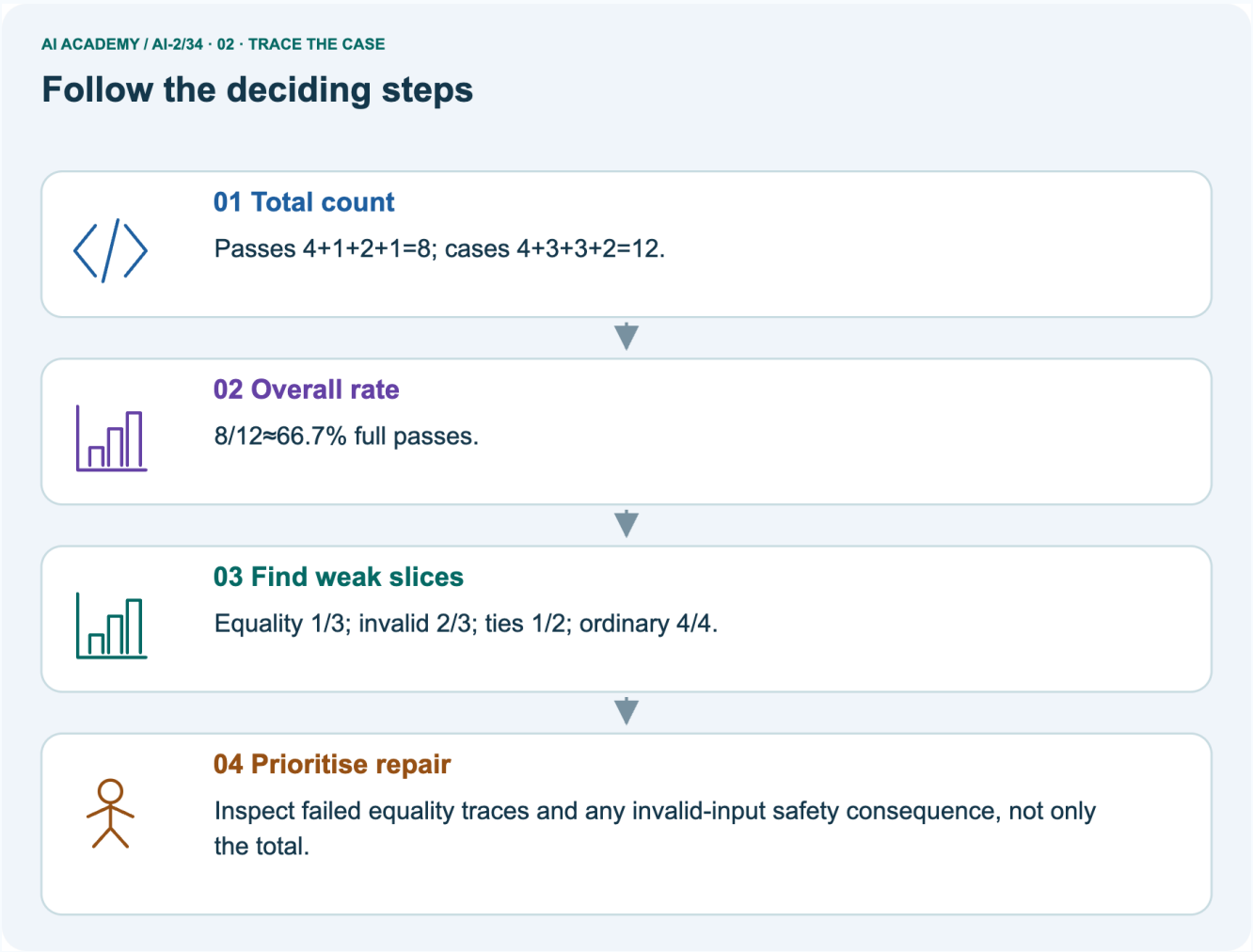
Read the separate twelve-case facts

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

Trace disjoint totals and weak conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Total count	Passes 4+1+2+1=8; cases 4+3+3+2=12.
2. Overall rate	8/12≈66.7% full passes.
3. Find weak slices	Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.
4. Prioritise repair	Inspect failed equality traces and any invalid-input safety consequence, not only the total.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Predict ordinary-only reporting

Change: Someone reports only the ordinary slice.

Prediction: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

Reason: Eight nonordinary cases and their failures have been omitted.

Explain what remains unmeasured

Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.

2. Run the same frozen recommender for all four.

3. Record expected, actual, score **evidence**, and issue type.

4. Do not average away one group’s repeated failure.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Exact explanation to model

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Explain the added worked example

Key terms

Term	Meaning
Evaluation set	Cases used to assess a specified version
Failure category	A type of scenario where an expected result is missed
Pass rate	Passing cases divided by executed cases
Coverage	The variety of requirements and conditions tested

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Test Diverse Users

The facts we will use

AI ACADEMY / AI-2/34

Disjoint evaluation categories

	Full passes	Cases run
Ordinary	4	4
Equality	1	3
Invalid	2	3
Ties	1	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Name five distinct categories for the chooser evaluation.

Expected reasoning and fading help

Teacher check: Ordinary selection, duration boundaries, invalid records, empty/no-match **input** and ties/order reversal.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an invented satisfaction score proves educational benefit.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Can these records establish performance for different user groups?

No. They describe **input**-case types. No user-group membership or group-specific outcomes were supplied.

Changed-case reasoning: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Write the question to explain: Compare and investigate.

Reveal after **prediction**s: Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Use the earlier changed question as optional later practice. Reserve Audit a new suite independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

15/16 pass, but no new condition was tested. The larger fraction is misleading as **evidence** of broader quality. Exit: The old case is now a regression/development case.

Document the fix and use fresh reserved cases for a new independent final check.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

User-condition traces: U1 selects P; U2 has no eligible option; U3 selects Q; U4 has no verified eligible option because Q has no supplied screen-reader check. Correct abstention can satisfy the contract while a user need remains unmet. These are paper traces, not interface or user-group measurements. Six-case suite: 5/6 overall and 0/1 ties; ten duplicate passes produce 15/16 without new coverage.

Separate picture: 8/12 overall; 4/4 ordinary does not describe all cases. Independent suite: 6/8, with one boundary and one tie failure; consequences are needed to order the repairs.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the supplied fictional P/Q/R catalogue and U1–U4 request cards. Trace fixed constraints, record correct no-match separately from an unmet need, and label the exercise a paper trace. No actual user-group or assistive-technology performance measurement is supplied.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful.

Neither a pass fraction nor an invented satisfaction score proves educational benefit.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: ordinary 2/2, empty 0/1, boundary 2/2 gives 4/5 overall; the empty case still fails.

Require the failure record and a fresh-case plan.

A convenient shortcut preserves meaning: Hiding the tie failure in an average conceals an unmet requirement. Preserve the failed **input**, expected result, actual result and version.

If you repair using it, classify it as development/regression **evidence** and reserve new final cases.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Audit a new suite independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the fresh eight-case suite

> Keep this key until after the independent attempt.

Selected question and key

Selected case: Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

6/8 pass, 2 fail; boundary has one failure and ties fail the only case. Report both and avoid treating 75 percent as production accuracy.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 · Hidden category weakness

A report says 8 of 10 tests pass, including 0 of 2 invalid-**input** cases. What is the important weakness?

Invalid-**input** handling fails both tested cases even though the overall pass fraction is 80 percent.

W2 · Distinct coverage

Name five distinct categories for the chooser evaluation.

Ordinary selection, duration boundaries, invalid records, empty/no-match **input** and ties/order reversal.

Use ordinary eligible selection, duration boundaries, schema-invalid records, empty or no-match **input**, and tied scores with reversed order. Each category checks a different requirement.

Include concrete **input**s and expected reports; five copies of one ordinary case do not provide this **coverage**.

W3 · Restore failures

A learner drops two failed cases and reports 8/8. What should be restored?

Report all 10 executed cases with 8 passes and 2 failures, category details and the tested version.

W5 · Protect final evidence

You fix the tie bug using the final tie case. What should the next evaluation report say?

The old case is now a regression/development case. Document the fix and use fresh reserved cases for a new independent final check.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Additional transfer practice question

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

Reasoned practice answer

Easy requests: 6/6; advanced: 1/4. Investigate advanced requests and inspect their failures. The combined 7/10 hides this difference.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A single overall score describes every user's experience.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Overall is about 73.3%; mystery is 90%; poetry is 40%. The total hides weaker poetry performance.

Investigate catalogue **coverage**, labels and scoring without assuming the rates identify the cause; small denominators limit conclusions.

Explanations for the additional workbook examples

W1 Read denominators

W2 Plan coverage

W3 Repair reporting

W4 Evaluate independently

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

6/8 pass, 2 fail; boundary has one failure and ties fail the only case. Report both and avoid treating 75 percent as production accuracy.

W5 Protect final evidence

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use a category-by-case grid. Extension: add a new case that tests a different failure **mechanism** rather than duplicating a passing case.

Independent W4 **evidence**: 6/8 pass, 2 fail; boundary has one failure and ties fail the only case. Report both and avoid treating 75 percent as production accuracy.

Record whether the learner needed no help, a hint or explanation. Fresh retry: ordinary 2/2, empty 0/1, boundary 2/2 gives 4/5 overall; the empty case still fails. Require the failure record and a fresh-case plan.

Plan the delayed check

Suggested return points in this level: ai-2/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 35: Improve and Challenge. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week35: Improve and Challenge. Preserve the failed cases, version and **evidence** limits when comparing a repair with the baseline.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Evaluate across varied cases

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Session:** Session 2 · Practise · 45 min

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> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Name five distinct categories for the chooser evaluation.

Reasoned practice answer: Ordinary selection, duration boundaries, invalid records, empty/no-match **input** and ties/order reversal.

Check every learner before discussion: Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an invented satisfaction score proves educational benefit.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Someone reports only the ordinary slice. Require a **prediction** and a reason before comparing. Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

Reason: Eight nonordinary cases and their failures have been omitted.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

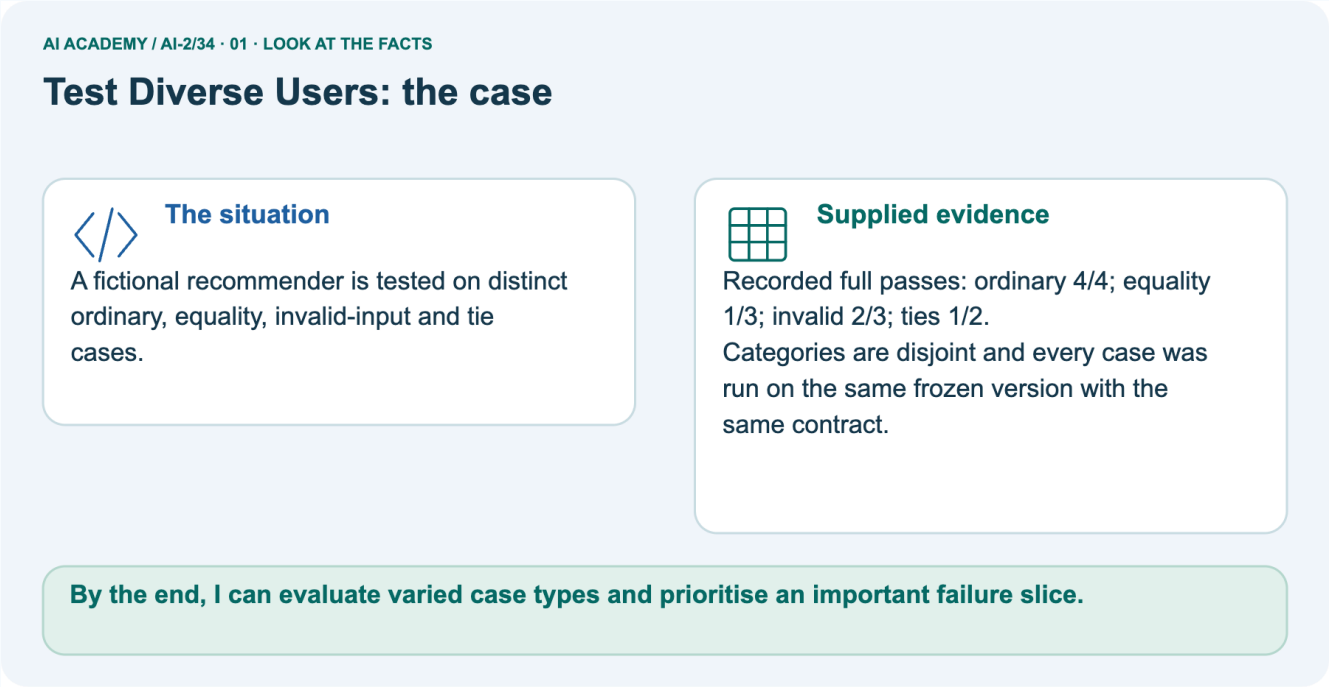
<details>
<summary>Reference: teach with the six pictures</summary>
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REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Test Diverse Users: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

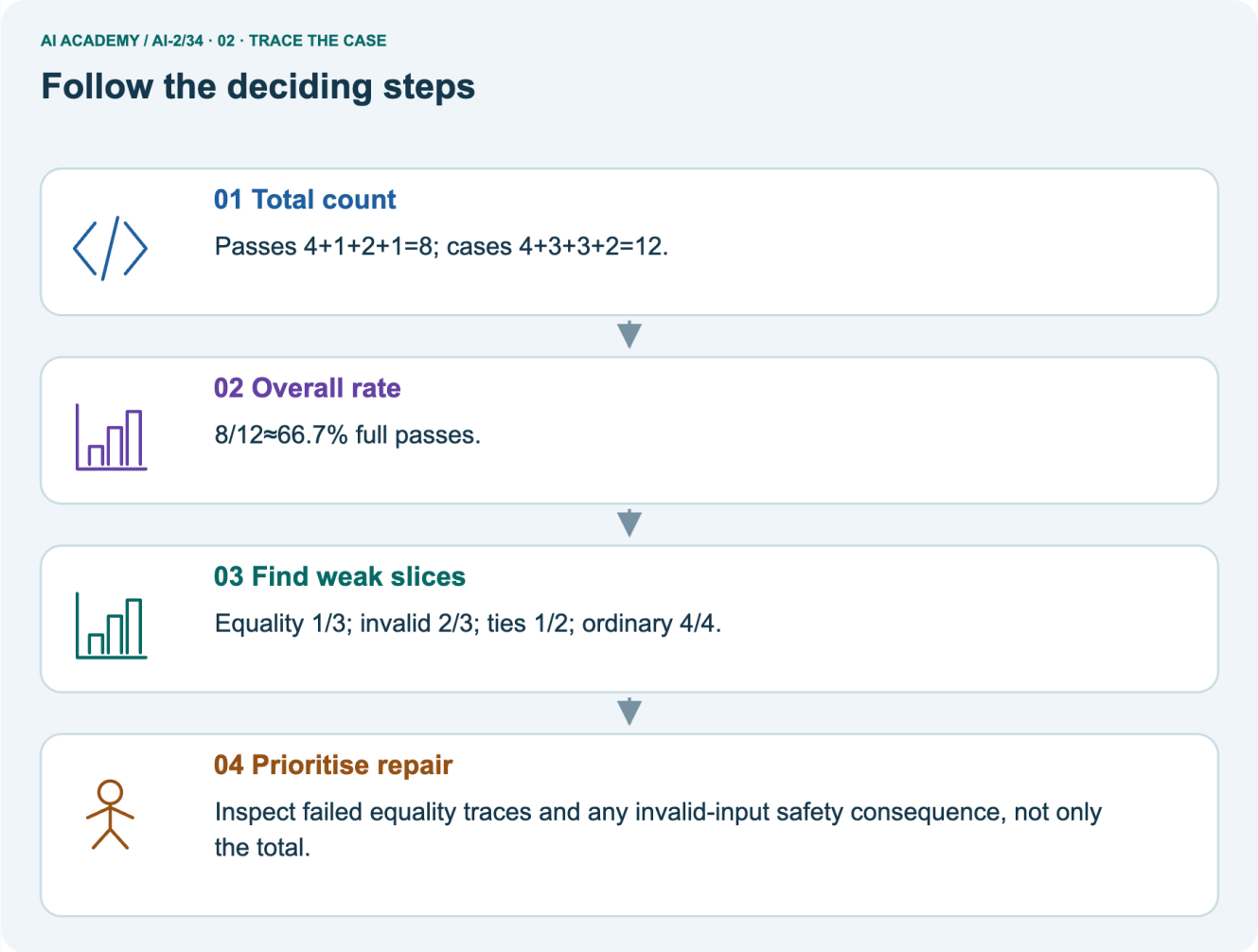
A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.
Supplied facts: Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



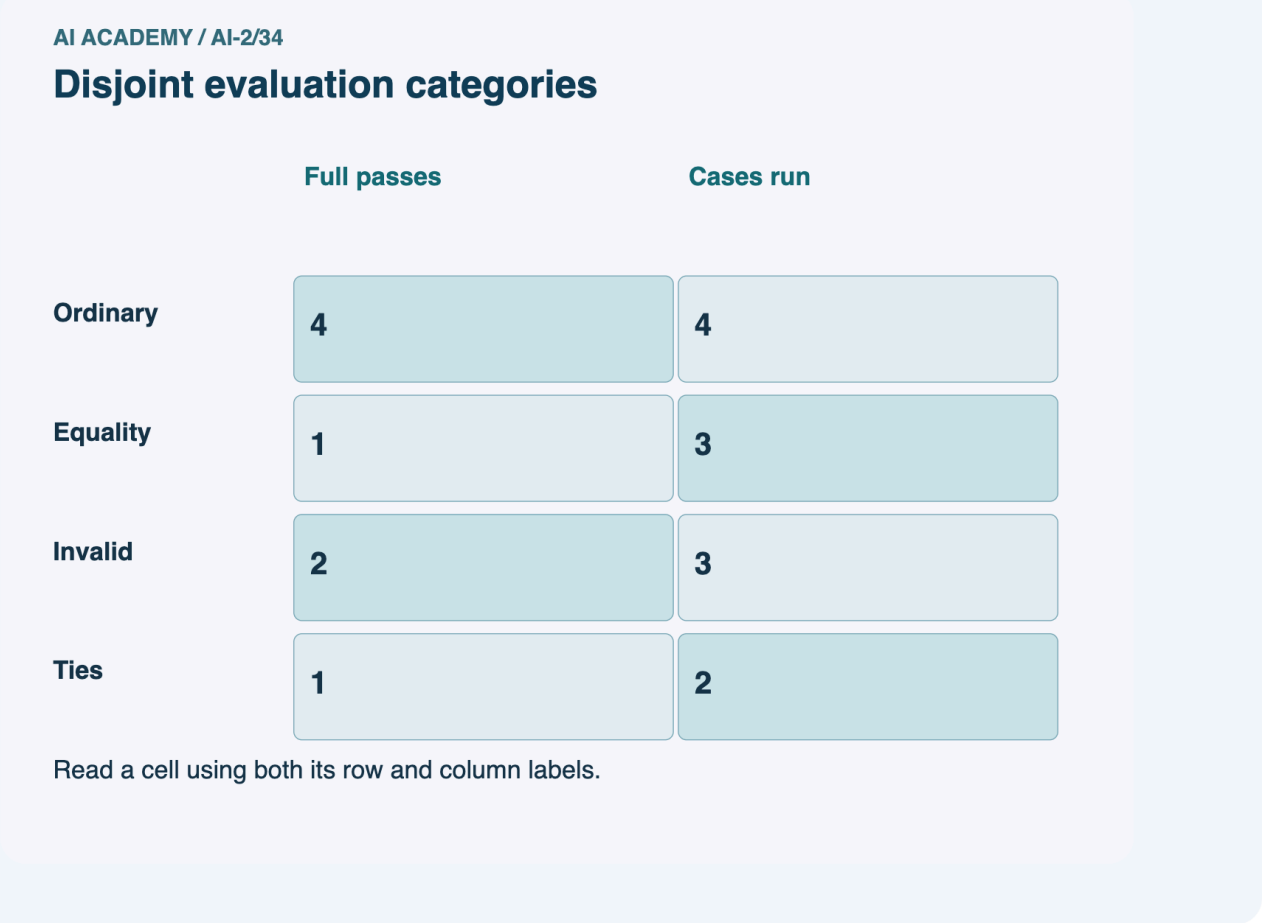
Follow the labelled picture with the matching explanation. Open the vector diagram

Total count: Passes 4+1+2+1=8; cases 4+3+3+2=12. → Overall rate: 8/12≈66.7% full passes. → Find weak slices: Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4. → Prioritise repair: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Disjoint evaluation categories



Follow the labelled picture with the matching explanation. Open the vector diagram

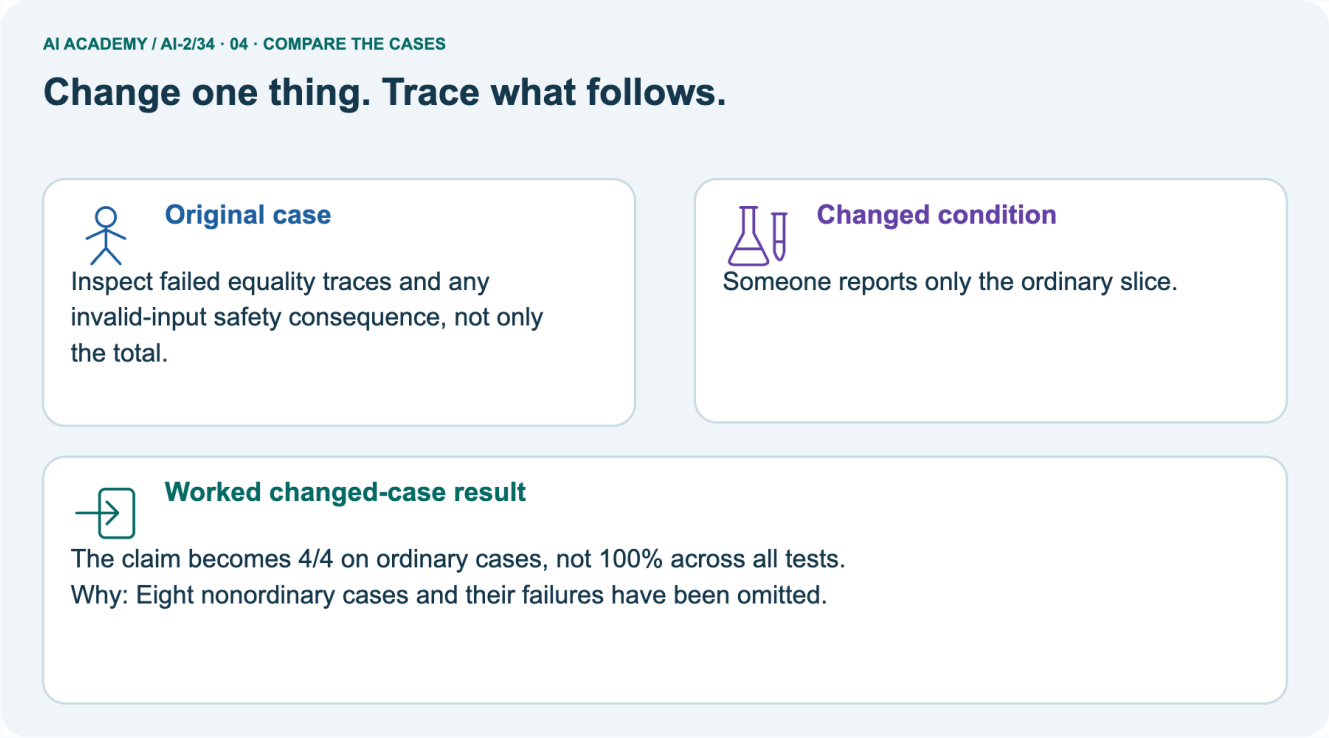
Disjoint evaluation categories. Ordinary: Full passes = 4, Cases run = 4. Equality: Full passes = 1, Cases run = 3. Invalid: Full passes = 2, Cases run = 3. Ties: Full passes = 1, Cases run = 2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total. Changed condition: Someone reports only the ordinary slice.

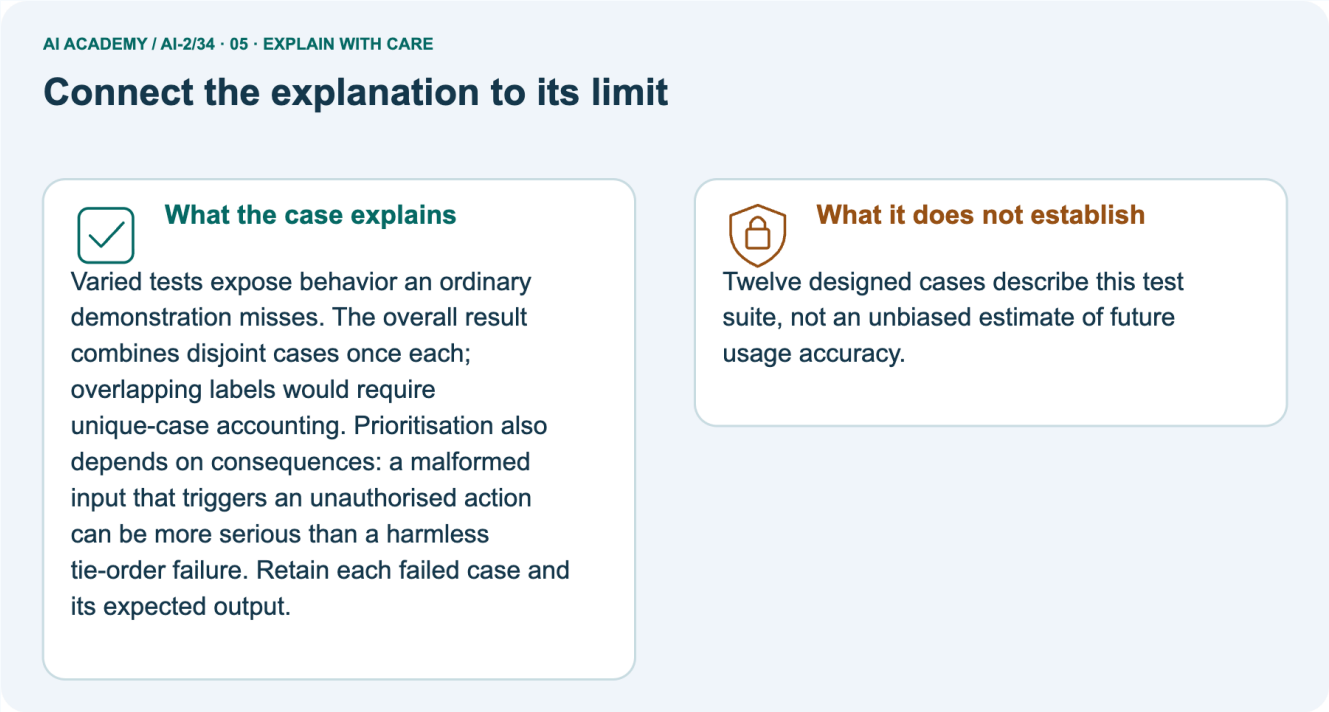
Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Reason: Eight nonordinary cases and their failures have been omitted.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Eight nonordinary cases and their failures have been omitted.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

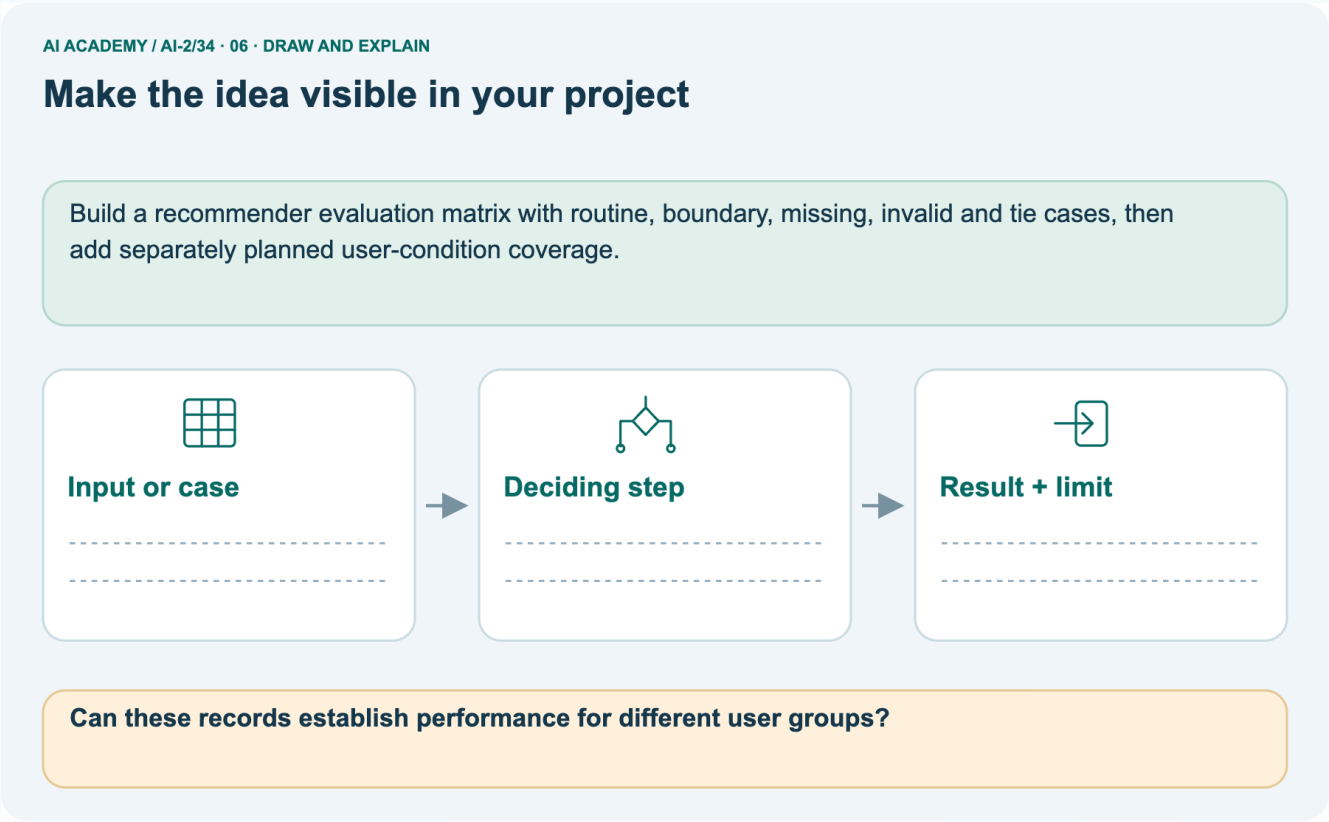
Limit: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**. Draw your **input**, deciding step, result and limit.

Supported question: Can these records establish performance for different user groups?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. They describe **input**-case types. No user-group membership or group-specific outcomes were supplied.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Test Diverse Users: the case

02. Follow the deciding steps

03. Disjoint evaluation categories

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.